BOLUTITO BABATUNDE

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EDUCATION

Carnegie Mellon University — Pittsburgh, PA
Doctor of Philosophy in Mechanical Engineering — August 2024
Master of Science in Mechanical Engineering — December 2022
Advisors: Dr. Rebecca E Taylor, Dr. Jonathan Cagan
Thesis: *A flexible framework for automating structural multilayer DNA origami designs*

Texas Tech University — Lubbock, TX
Bachelor of Science in Mechanical Engineering — May 2019
Minor: Computer Science, Mathematics

SKILLS

Languages: Python, C++ (OpenGL), Bash, R, Perl
Frameworks: NumPy, SciPy, pandas, PyTorch, AWS, oxDNA MD Simulation, Agarose Gel Electrophoresis, Atomic Force Microscopy, DNA Nanotechnology
Software & Platforms: SLURM, Git, GATK, Clair3, PLINK2, Modkit, methyKit, bamql, coloc, STAR, fastp, Qualimap, IGV, tensorQTL, bedtools, samtools, FitHiC, Mustache, DiffHiC, Dorado, WhatsHap, HiC-Pro, edgeR, DESeq2, TOBIAS, HOMER, MACS3, ChimeraX, CaDNAno, Scadnano

PROFESSIONAL DEVELOPMENT

AAI Introductory Course in Immunology (Parts 1 & 2) — Los Angeles, CA
American Association of Immunologists (AAI) — July 2025

RESEARCH EXPERIENCE

La Jolla Institute for Immunology and Allergy — San Diego, CA
POSTDOCTORAL RESEARCHER, Ay Lab — October 2024 - Present
NIH T32 Institutional Training Grant Fellow (October 2024 - July 2026)

- Established an Oxford Nanopore pipeline (Dorado, modkit, Clair3, tensorQTL) to map genetic variants that shape CpG modifications (5mC & 5hmC) in human nCD4+ T cells from 41 donors (~30x coverage)
- Integrated matched ATAC-seq and RNA-seq with 5hmC/5mC QTL maps to evaluate downstream chromatin and expression changes, cross-referencing results with eQTL and hQTL databases to refine causal regulatory variants
- Adapted a single-molecule nano-NOMe-seq pipeline for nCD4+ T cells across 19 donors to profile GpC methylation (chromatin accessibility) and CpG modifications on individual reads, leveraging phased heterozygous variants to analyze allele-specific chromatin states, TF footprints, and QTL effects
- Led the computational analyses for a collaboration with Dr. Vijay K. Kuchroo's lab comparing mouse pathogenic and non-pathogenic Th17 cells with Th0, Th1 and Treg lineages, linking chromatin loop changes (HiC-Pro, Mustache, FitHiC, edgeR) to differential gene expression and chromatin accessibility (ATAC-seq peaks)
- Mapped age-associated 5mC differential methylation in human CD8+ T cell subsets (naïve, TEM, TEMRA) via Oxford Nanopore sequencing, connecting epigenetic shifts to gene and transposable element expression
- Supported a collaboration with Dr. Debattama R. Sen's lab on how Dr1 enhancer dysregulation shapes exhausted CD8+ T cell states in a mouse tumor model, contributing ATAC-seq, Hi-C loop and TF footprinting analysis

Carnegie Mellon University — Pittsburgh, PA
GRADUATE RESEARCHER — August 2019 - August 2024
Investigating a flexible framework for automating multilayer DNA origami design

- Developing a computational casting technique to automate solid and hollow DNA origami designs by casting a long single-stranded DNA (ssDNA) around or within a triangular mesh leveraging generative optimization strategies (i.e shape annealing)
- Integrating an adjustable ssDNA axis orientation feature to enhance design capabilities, influence structural density, uniformity, and generating novel routing patterns
- Created custom Python algorithms to convert generated designs into formats compatible with standard bioinformatic tools, scadnano, caDNAno and oxDNA
- Developed a custom Python stapling algorithm to efficiently automate routing of hundreds of short oligonucleotides (or ssDNAs) to cinch long ssDNA routing patterns generated by computational casting technique
- Supervising an in-house lab technician to manufacture and characterize designs using gel-electrophoresis and collaborating with an Assistant Professor at University of Pittsburgh to ensure formation with transmission electron microscopy (TEM)
- Partnering with a PhD student to create new lab website and maintain Ubuntu 20.04 lab server and Amazon Web Services (AWS) Elastic Compute Cloud (EC2) instances for lab members

Los Alamos National Laboratory Los Alamos, NM
ATHENA SCHOLAR GRADUATE INTERN May 2019 - August 2019

- Constructed a functional prototype of a temperature controller connected to a chamber for heating specimens using Programmable Logic Controller (PLC) programmed with LABVIEW (systems engineering software) for remote system operation
- Leveraged Robotic Operating System (ROS) and MoveIt to program a robotic arm to grab object for heating

ACADEMIC PROJECTS

(LTI-11685) Introduction to Deep Learning Pittsburgh, PA
PROTEIN LANGUAGE MODELING: CODING LIFE'S CODE August 2023 - December 2023

- Teamed up with master's student to modify the PyTorch-based SPOT-1D-LM model by incorporating ProteinBERT embeddings and integrating additional ResNet layers for single sequence analysis to investigate protein structure-function relationship
- Utilized modified SPOT-1D-LM to classify eight-state labels for protein secondary structure (SS8), achieving comparable accuracy, including on challenging Neff1-2020 set with no homologs (69.07%)
- Conducted an ablation study to eliminate overfitting by introducing weight decay and L1 regularization, leading to improved performance and validation loss stability from 5th epoch onward

(MEG-24787) Machine Learning and AI for Engineers Pittsburgh, PA
CLASSIFYING SCAFFOLD ROUTINGS OF DNA ORIGAMI DESIGNS August 2022 - December 2022

- Implemented a custom convolutional neural network (CNN) utilizing PyTorch for gap detection in cross-sectional view images of hollow designs produced by computational casting technique
- Partnered with PhD student to develop a custom CNN, achieving 73% accuracy for encapsulated ssDNA designs, outperforming VGG16 (86%) and ResNet50 (78%) showed increased loss after 20 epochs
- Improved accuracy to 92% by reclassifying dataset into three classes: large gap, small gap, and encapsulated

PUBLICATIONS

- **Babatunde, Bolutito**; Cagan, Jonathan; Taylor, Rebecca E. (2024). "An Improved Shape Annealing Algorithm for the Generation of Coated Deoxyribonucleic Acid Origami Nanostructures." ASME Journal of Mechanical Design. 146, no. 5: 051708. https://doi.org/10.1115/1.4064242 **(Special Issue of the ASME Journal of Mechanical Design featuring top papers from IDETC 2023)**
- **Babatunde, Bolutito**; Arias, Sebastian D.; Cagan, Jonathan; Taylor, Rebecca E. (2021). "Generating DNA Origami Nanostructures through Shape Annealing." Applied Sciences. 11, no. 7: 2950. https://doi.org/10.3390/app11072950

CONFERENCES / PRESENTATIONS

- **Babatunde, Bolutito**; Dasgupta, Nirmalya; Chakraborty, Abhijit; Wang, Rosan; Amariuta, Tiffany; Ay, Ferhat (20-24 October 2026). "5-Hydroxymethylcytosine QTL Mapping in Human nCD4+ T Cells using ONT and Nano-NOMe-seq Long-Read Sequencing." American Society of Human Genetics (ASHG) 2026 Annual Meeting, Palais des Congrès de Montréal, Montréal, QC, Canada
- **Babatunde, Bolutito**; Wang, Rosan; Dasgupta, Nirmalya; Chakraborty, Abhijit; Ay, Ferhat (12-16 July 2026). "5-Hydroxymethylcytosine QTL Mapping in Human nCD4+ T Cells via Long-Read Sequencing." 34th Conference on Intelligent Systems for Molecular Biology (ISMB), Washington Hilton, Washington, DC, USA
- Dasgupta, Nirmalya; Chakraborty, Abhijit; **Babatunde, Bolutito**; Ay, Ferhat (6 November 2025). "Oxford Nanopore DNA and RNA sequencing projects from the Ay Lab." Invited seminar with Oxford Nanopore Technologies, La Jolla Institute for Immunology, La Jolla, CA, USA
- **Babatunde, Bolutito**; Wang, Rosan; Dasgupta, Nirmalya; Chakraborty, Abhijit; Ay, Ferhat (19-21 March 2025). "Investigating 5hmC enhancer function using short and long-read sequencing." 2025 LJI Scientific Retreat, UCLA Lake Arrowhead Lodge, Lake Arrowhead, CA, USA
- **Babatunde, Bolutito**; Cagan, Jonathan; Taylor, Rebecca E. (20-23 August 2023). "An Improved Shape Annealing Algorithm for the Generation of Coated DNA Origami Nanostructures." International Design Engineering Technical Conferences and Computers and Information in Engineering Conference (IDETC/CIE 2023), Boston Park Plaza, Boston, MA, USA
- **Babatunde, Bolutito**; Cagan, Jonathan; Taylor, Rebecca E. (8-12 August 2022). "A Refined Shape Annealing Algorithm for the Optimal Generation of DNA Origami Designs." DNA28: The 28th International Conference on DNA Computing and Molecular Programming (Track C), University of New Mexico, Albuquerque, NM, USA
- **Babatunde, Bolutito**; Cagan, Jonathan; Taylor, Rebecca E. (4 March 2022). "A Formal Automated Approach for Controlling the Wall Thickness of Coated DNA Origami Designs." 2022 PhD Research Symposium, Carnegie Mellon University, Pittsburgh, PA, USA. **(Best poster award in MicroNano Research)**
- **Babatunde, Bolutito**; Arias, Sebastian D.; Cagan, Jonathan; Taylor, Rebecca E. (12-15 April 2021). "A Formal Approach for Automated Generation of DNA Origami Designs." FNANO 2021: 18th Annual Conference Foundations of Nanoscience (Computational Tools), Virtual